

2.2 11, 15, 20, 27, 31, 34

11.

首先证明 $X = A^{-1}B$ 是 $AX = B$ 的解

$$A(A^{-1}B) = (AA^{-1})B = I_n B = B \quad \therefore (A-AX)(A-AX)^{-1}$$

其次证明解唯一 - 对于所有 X

$$AX = B \rightarrow A^{-1}AX = A^{-1}B$$

$$\rightarrow I_n X = A^{-1}B \rightarrow X = A^{-1}B$$

$\therefore AX = B$ 有唯一解 $A^{-1}B$

15. 可令 $D = C^{-1}B^{-1}A^{-1}$, 此时

$$\begin{aligned} (ABC)D &= (ABC)(C^{-1}B^{-1}A^{-1}) \\ &= A\underline{B(C \cdot C^{-1})}B^{-1}A^{-1} \\ &= A(B \cdot I_n B^{-1})A^{-1} \\ &= AA^{-1} = I_n \end{aligned}$$

同理

$$\begin{aligned} D(ABC) &= (C^{-1}B^{-1}A^{-1})(ABC) \\ &= C^{-1}B^{-1}(A^{-1}A)B \cdot C \\ &= I_n \end{aligned}$$

$\therefore ABC$ 也可逆

20. a.

$$\therefore (A-AX)^{-1} = X^{-1}B$$

$$\therefore (A-AX)(A-AX)^{-1} = (A-AX) \cdot X^{-1} \cdot B = I$$

$\therefore A-AX$ 为 $n \times n$ 矩阵
 X^{-1} 为 $n \times n$ 矩阵.

$\therefore (A-AX) \cdot X^{-1}$ 为 $n \times n$ 矩阵

$\therefore$ 由定理 8,

B 为可逆矩阵

b

$\therefore B$ 为可逆矩阵, X 为可逆矩阵 $\therefore X^{-1}B$ 可逆

$$\begin{aligned} \therefore A-AX &= (X^{-1}B)^{-1} \\ &= B^{-1}(X^{-1})^{-1} = B^{-1}X \end{aligned}$$

$$\therefore A = \overset{-1}{B}X + AX$$

$$A = (B^{-1} + A)X \quad (a)$$

对 (a) 令 X^{-1} 考虑以下式子

$$AX^{-1} = B^{-1} + A$$

$\therefore A$ 可逆, X^{-1} 可逆, $\therefore AX^{-1}$ 可逆 $\therefore B^{-1} + A$

可逆, $\therefore X = (B^{-1} + A)^{-1}A$

27. a. 2.1节方程(1):

$$A(Bx) = (AB)x$$

$$A(Bx) = \begin{bmatrix} \text{row}_1(A) \\ \vdots \\ \text{row}_i(A) \\ \vdots \\ \text{row}_n(A) \end{bmatrix} (Bx)_{21 \times 24}$$

$$= \begin{bmatrix} \text{row}_1(A) \cdot B \\ \vdots \\ \text{row}_i(A) \cdot B \\ \vdots \\ \text{row}_n(A) \cdot B \end{bmatrix} x$$

$$= (AB)x = \begin{bmatrix} \text{row}_1(AB) \\ \vdots \\ \text{row}_n(AB) \end{bmatrix} x$$

$$\therefore \text{row}_i(A) \cdot B = \text{row}_i(AB)$$

$$\text{令 } A = I, B = A$$

$$\therefore \text{row}_i(I) \cdot A = \text{row}_i(A)$$

其中 i 可以扩展为 Z .

b 使用(a)的结论.

$$\begin{bmatrix} \text{row}_2(A) \\ \text{row}_1(A) \\ \text{row}_3(A) \\ \vdots \\ \text{row}_n(A) \end{bmatrix} = \begin{bmatrix} \text{row}_2(I) \cdot A \\ \text{row}_1(I) \cdot A \\ \text{row}_3(I) \cdot A \\ \vdots \\ \text{row}_n(I) \cdot A \end{bmatrix} = \begin{bmatrix} \text{row}_2(I) \\ \text{row}_1(I) \\ \text{row}_3(I) \\ \vdots \\ \text{row}_n(I) \end{bmatrix} \cdot A = EA$$

c. 同样使用(a)中结论.

$$c. \begin{bmatrix} \text{row}_1(A) \\ \text{row}_2(A) \\ \text{row}_3(A) \\ \vdots \\ \text{row}_n(A) \end{bmatrix} = \begin{bmatrix} \text{row}_1(I) \cdot A \\ \text{row}_2(I) \cdot A \\ \text{row}_3(I) \cdot A \\ \vdots \\ \text{row}_n(I) \cdot A \end{bmatrix}$$

$$= \begin{bmatrix} \text{row}_1(I) \\ \text{row}_2(I) \\ \text{row}_3(I) \\ \vdots \\ \text{row}_n(I) \end{bmatrix} A = EA$$

$$[A \ I] =$$

$$3 | \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{bmatrix}$$

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{bmatrix}$$

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{bmatrix}$$

$$\rightsquigarrow \begin{bmatrix} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

$$= [I \ A^{-1}]$$

$$\therefore A^{-1} = \begin{bmatrix} 8 & 3 & 1 \\ 10 & 4 & 1 \\ \frac{7}{2} & \frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

34.
guess: $B = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 & \cdots & 0 \\ 0 & -\frac{1}{3} & \frac{1}{3} & \cdots & 0 \\ 0 & 0 & -\frac{1}{4} & \frac{1}{4} & \vdots \\ \vdots & & & & -\frac{1}{n} & \frac{1}{n} \\ & & & & & - \end{bmatrix}$

$$\begin{aligned} \therefore AB_i &= A\left(\frac{1}{i}e_i - \frac{1}{i+1}e_{i+1}\right) \\ &= \frac{1}{i}a_i - \frac{1}{i+1}a_{i+1} \\ &= e_i \end{aligned}$$

$$\therefore AB = I$$

$\therefore$ 定理 8 可知, B 是 A 的逆.

2.3 7, 10, 12, 26, 33 10. n

$$7. \begin{bmatrix} -1 & -3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & -3 & 0 & 1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & -1 & 2 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & -3 & 0 & 1 \\ 0 & -4 & 8 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

有4个主元位置, 可逆.

$$10. \begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 6 & 4 & 2 & 8 & -8 \\ 7 & 5 & 3 & 10 & 9 \\ 9 & 6 & 4 & -9 & -5 \\ 8 & 5 & 2 & 11 & 4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 0.4 & 0.8 & -0.4 & -18.8 \\ 0 & 0.8 & 1.6 & 0.2 & -3.6 \\ 0 & 0.6 & 2.2 & -21.6 & -21.2 \\ 0 & 0.2 & 0.4 & -0.2 & -10.4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 0.4 & 0.8 & -0.4 & -18.8 \\ 0 & 0 & 0 & 1 & 34 \\ 0 & 0 & 1 & -21 & 7 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 3 & 1 & 7 & 9 \\ 0 & 0.4 & 0.8 & -0.4 & -18.8 \\ 0 & 0 & 1 & -21 & 7 \\ 0 & 0 & 0 & 1 & 34 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

有5个主元位置, 可逆.

12. a. ✓

b. ✓

c. ✓

d. ✓

e. ✓

26. A各列线性无关. 由定理8.

A可逆, 则 A^T 可逆. (定理6)

则由定理8, A^T 各列生成 R^n

$$33. T \text{ 的矩阵为 } T = \begin{bmatrix} -5 & 9 \\ 4 & -7 \end{bmatrix}$$

$$\because \det A = 35 - 36 = -1 \neq 0$$

T可逆.

$$T^{-1} = \frac{1}{-1} \begin{bmatrix} -7 & -9 \\ -4 & -5 \end{bmatrix} = \begin{bmatrix} 7 & 9 \\ 4 & 5 \end{bmatrix}$$

$$\therefore T^{-1}(x_1, x_2) = (7x_1 + 9x_2, 4x_1 + 5x_2)$$

2.4. 7. 15

$$\therefore \begin{cases} X = A_{21} A_{11}^{-1} \\ Y = A_{11}^{-1} A_{12} \end{cases}$$

$$7. \begin{bmatrix} X & 0 & 0 \\ Y & 0 & I \end{bmatrix} \begin{bmatrix} A & Z \\ 0 & 0 \\ B & I \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

$$\text{左边} = \begin{bmatrix} XA & XZ \\ YA+B & YZ+I \end{bmatrix}$$

$$\begin{cases} XA = I \\ XZ = 0 \\ YA+B = 0 \\ YZ+I = I \end{cases} \quad \begin{cases} X = A^{-1} \text{ (A可逆)} \\ Z = 0 \\ Y = -BA^{-1} \end{cases}$$

$$15. \begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix}$$

$$= \begin{bmatrix} A_{11} & 0 \\ XA_{11} & S \end{bmatrix} \begin{bmatrix} I & Y \\ 0 & I \end{bmatrix}$$

$$= \begin{bmatrix} A_{11} & A_{11}Y \\ XA_{11} & XA_{11}Y+S \end{bmatrix}$$

$$\begin{cases} A_{11}Y = A_{12} \\ XA_{11} = A_{21} \\ XA_{11}Y+S = A_{22} \end{cases}$$

$$\begin{cases} Y = A_{11}^{-1} A_{12} \\ X = A_{21} \cdot A_{11}^{-1} \end{cases}$$

$$\text{验证: } A_{21} \cdot A_{11}^{-1} \cdot A_{11} \cdot A_{11}^{-1} A_{12} + A_{22} - A_{21} A_{11}^{-1} A_{12} = A_{22}$$